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Modelling smectite illitization in earthen barriers of buried radioactive wastes

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The textural characteristics of engineered clay barriers of deeply buried radioactive waste repositories are likely to change in response to hydro-thermo-mechanical stresses during the 10 000-year service and risk analysis period. Expected textural changes stem from a combination of processes, the most significant of which is smectite illitization. To pose significant risks of radionuclide release from buried waste canisters, illitization must, as a minimum, produce barrier textural changes that are significant within the timeframe in which radioactive activity is still significant. Besides, the canister must be damaged, principally through corrosion by substance-laden infiltrating moisture. In the current paper, simulations of three scenarios of illitization of a clay barrier that is initially 100% smectite under a temperature regime that does not exceed 150°C indicate that the accumulation of potassium ions (K^+) borne by infiltrating water close to the waste canister is likely to be insignificant within the first 1000 years of waste entombment. Some K^+ are extracted from the infiltrating moisture to feed the illitization reaction which is herein assumed to occur with an activation energy of 1.17×10^{-5} J/mole.

Keywords: Radioactive waste; Barrier material; Waste containment; Coupled transport; Smectite illitization

1. Introduction

Numerical models for analysis of the potential performance of repositories within the typical 10 000-year service and risk analysis period comprise source terms for estimation of radionuclide release rates from earthen barrier-encapsulated waste packages. It is often tacitly (and incorrectly) assumed that barrier texture-dependent properties such as diffusion coefficient, permeability, specific heat and thermal conductivity will be constant during the service life of the repository barrier. This assumption is a convenient circumvention of the uncertainties associated with predicting both the magnitudes and the rates of temporal changes in the clayey barrier mineralogy and texture. Inyang (1994) described this problem and proposed the Weibull model as the general numerical framework for its treatment. In this introduction section, general transport and textural change mechanisms that impact upon barrier performance are described prior to analysis of illitization, which is the primary textural change mechanism covered in this analysis. The general discussion provides a context for the role of illitization.

There are some performance factors that apply to deeply buried barriers but are largely insignificant with respect to near-surface barriers. For near-surface barriers, textural parameters discussed by Daniels *et al.* (2003), Reddi and Inyang (2000), Chen *et al.* (2000), Inyang and Tumay (1995), Kargbo *et al.* (1993), Fleming and Inyang (1995), Fang *et al.* (1997) and Inyang *et al.* (1998a,b) that are significant to contaminant transport are more influenced by a broader range of physico-chemical and biological processes. Near-surface environments have greater cycles of reversals in temperature and moisture condition than the deep environment in which radioactive waste barriers are entombed. On the other hand, greater heat and pressure at depth relative to shallow soil environments can 'cook' barriers to new mineralogical composition. Being that they are designed to perform for greater service lives (often up to 10 000 years), than near-surface containment system design lives, barriers in deep repositories provide greater opportunity for slow physico-chemical processes to produce cumulative effects that may be significant.

Several numerical models (Seitz *et al.* 1987, Soudek *et al.* 1983, Carnahan 1982) have been developed for estimating the rates of transport of radionuclides, moisture and heat through clayey barriers of radioactive waste repositories. The flow rates of both mass and energy through any barrier depend primarily on contaminant physico-chemical characteristics, continuity or transience of flow coupling, magnitudes of differences in mass and energy potentials across the barrier, and the dimensions and

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textural characteristics of the barrier. Sodium montmorillonite can be used alone or in combination with crushed basalt or graded silica sand as the barrier material. Considering that there is frequent references in literature to montmorillonite, smectite and bentonite as contaminant barrier materials, sometimes interchangeably, it is important to indicate the differences among them. Sodium montmorillonite, to which reference is made in the current paper, is a pure mineral that belongs to the smectite group of minerals. On the other hand, bentonite is a commercial barrier material comprising montmorillonite in high proportion with impurities. It is often mined and used in its natural form. These low permeability materials serve as overpack for the waste canisters so that access by the corrosive groundwater will be minimized. Furthermore, the outward flow potential of radionuclides is expected to be minimal owing to the high swelling capacity of pure and amended Na-bentonite. However, this observation is based on the assumption that phenomena such as those summarized in table 1 will not induce significant changes in the mineralogy of the barrier material.

Some textural changes that may accompany textural changes in barrier materials can only occur after prolonged physico-chemical reactions that may span time intervals greater than 500 years. The implication is that rate constants obtained in short-term laboratory-based experiments may be hopelessly inadequate as the basis for long-term predictions. Hydrothermal reactions at moderate temperature regimes (100–130°C) fit into this category. The kinetics of hydrothermal alteration of smectite barrier materials to illite during the service life of radioactive waste repositories has attracted considerable scientific interest during the past 20–25 years. While Eberl and Hower (1976) assumed that smectite illitization follows first order reaction pathway, Whitney (1990) performed laboratory-based experiments on Wyoming bentonite to estimate illitization rates at a pressure of 100 MPa, temperature of 250°C and solid:liquid ratios ranging from 1:1 to 20:1. An analysis of the data presented by Whitney (1990) shows that with increase in time, the forward reaction constant for illitization of smectite under the test conditions decreased significantly. The rate constants computed using experimental data of Whitney (1990) are presented in table 2. Thus, progressively higher activation energies may be required to sustain the chemical reaction. In essence, within the context of earthen barrier performance analysis, the utility of most data obtained by measurement in the laboratory may rest more on their use in supporting or discounting the relevance of reaction mechanisms than on estimating the potential magnitudes of the associated barrier textural changes.

Table 1. Potential effects of various physico-chemical processes on transport properties of clayey barriers of waste repositories

Alteration mechanism	Inducing conditions	Primary effect on clayey barrier properties
Dehydration	Temperature increase to levels above 100°C	Decrease in thermal conductivity
Infiltration of moisture and electrolytes	High water pressure and high permeability of host rock	Increase in thermal conductivity and increase in radionuclide diffusivity
Radiation damage	Waste canister failure and high dose rate of ionizing radiation	Edging of crystalline minerals and possible reduction in swelling potential
Dissolution and leaching	Moderate to high temperature (greater than 50°C) and availability of moisture	Increases in diffusion coefficient and permeability; and decrease in retardation coefficient for radionuclides
Precipitation	High concentration of salts in infiltrating groundwater; and suitable Eh/pH and temperature conditions	Possible increase in thermal conductivity; decrease in diffusion coefficient; an increase in retardation coefficient for radionuclides
Illitization	Intense hydrothermal conditions over long time periods; and availability of K^+ and Al^{3+}	Increase in diffusion coefficient; and decrease in swelling potential and retardation coefficient for radionuclides

Table 2. Forward reaction rate constants for the conversion of smectite to illite at a pressure of 100 MPa and temperature of 250°C (data from Whitney, 1990)

		C_t (%)			K_T (number/day)		
C_o %	S:L ratio	1 day	14 days	60 days	1 day	14 days	60 days
100	1:1	88	71	57	1.3×10^{-1}	2.4×10^{-2}	9.4×10^{-3}
100	5:1	85	69	58	1.6×10^{-1}	2.6×10^{-2}	9.1×10^{-3}
100	10:1	85	78	62	1.6×10^{-1}	1.8×10^{-2}	8.0×10^{-3}
100	20:1	95	89	62	5.1×10^{-2}	8.3×10^{-3}	3.3×10^{-3}

Clayey barriers in deep radioactive waste repositories are subjected to environmental conditions that are different from those that characterize the surficial environment. In the former, temperatures and confining pressures are high (in the ranges of 50 to 240°C and 9–30MPa respectively, during the repository service life); direct meteorological impacts are minimal; fluctuations in stresses and ambient temperature are less frequent; and except for the large shear deformations that can result from seismic events, damage are relatively slow. Under this new set of conditions, surface-mixed barrier materials will undergo physical and mineralogical changes towards the restoration of physical and mineralogical equilibrium. The primary changes that are likely are summarized in table 1.

The use of natural analogues is then a credible option in the light of the constraints on experimental simulation of the influence of long reaction durations on the processes that control barrier texture and associated transport characteristics. Analogues such as diagenesis and low-grade metamorphism of clayey sediments in depositional basins are useful for estimating the long-term changes that are likely to occur in clayey barrier materials in deep repositories. In the current paper, the effects of smectite to illite conversion on the access of K^+ to the radioactive waste canister are modelled. Potassium is used as the model ion because its entry into the barrier from surrounding groundwater can result in illitization and affect the travel of potentially corrosive substances to the surface of the waste-containing canister through the process depicted in figure 1. The principal outcome of these numerical experiments is that some conversion takes place near the waste container over a relatively short time span but then the illitzation process slows, first owiung to lack of moisture and subsequently owing to a decline in heat. Consequently, the overall effect of the conversion on fluid transport through the barrier system should be expected to be minimal.

2. Physico-chemical conditions for the illitization of barrier smectite

The primary type of hydrothermal alteration of clayey barrier material that may occur in repositories is illitization of smectite, i.e. the conversion of smectite to illite. In consistence with the aluminium-conserved reaction derived by Boles and Franks (1979)

$$3.93\ K^+ + 1.57\ \text{Smectite} \rightarrow \text{Illite} + \text{other byproducts} \quad (1)$$

Equation (1) can be further detailed as:

$$\begin{aligned} &Ex_x[Al_{2-x}Mg_x][Si_4]O_{10}(OH)_2 \rightarrow \\ &\quad K_{1-x}[Al_2][Al_{1-x}Si_{3+x}]O_{10}(OH)_2 \end{aligned} \quad (2)$$

The term [Ex] represents exchangeable cations which are usually Na ions in the smectite clay (montmorillonite) of repository seals. Illitization is promoted by high confining pressure, intense hydrothermal conditions and the availability of sources for both K^+ and Al^{3+}. Potassium ion (K^+) is needed as an exchangeable ion whereas Al^{3+} replaces Si^{4+} in the tetrahedral unit cell. Both K^+ and Al^{3+} can be derived from igneous and metamorphic

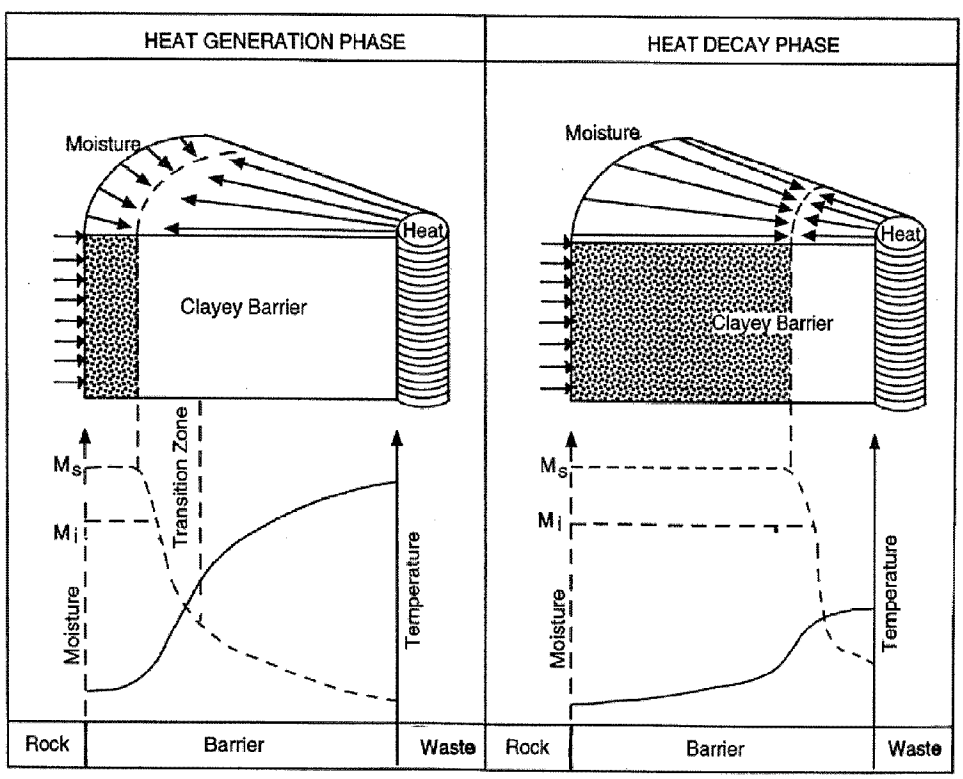

Figure 1. Stages during the dehydration and infiltration of clayey barriers of radioactive waste repositories.

host rocks. The dissolution of K-feldspars and mica produces K^+ and Al^{3+} respectively. These cations can diffuse into the clayey barrier from the pore solution of the host rock. Illite has a layer charge of −0.67 charges/unit cell in comparison with −0.33 charges/unit cell for montmorillonite. Thus, illitization creates additional negative charges. The implication is that very strong bonding develops between the clayey layers and the interlayer K^+. Experimental investigation (Bethke *et al.* 1986) indicates that segregation of clay particles develops due to partial conversion of smectite to illite. In such laboratory-based experiments, a redistribution of pores from smaller to larger sizes usually accompanies a collapse of interlayer spacing to about 10 Å (Å = Ångstroms). There is a concern that if illitization occurs significantly in clayey barriers of repositories, the transport of radionuclides through the barrier can be enhanced.

The waning of the illitization reaction implies that very long time intervals (in the order of millions of years) may be required for full conversion of smectite barriers to illitic barriers in repositories. Chapman *et al.* (1984) estimated that even at adequately high temperatures, more than 10^7 years may be required for full conversion. It is expected that temperature in full-scale repositories will exceed 150°C only during the first few years of waste burial. Illitization is generally insignificant below 150°C. In the current paper, the significance of illitization is not placed on full conversion but on the zonation of the barrier into the saturated zone and desiccated zone on either side of the illitization rim, however thin it may be.

3. Spatio-temporal sequence of barrier zonation

Depending on the size of the waste canister and the type of radioactive waste present, the temperature at the canister/barrier interface as illustrated in figure 1, may exceed 150°C within 10 years owing to radioactive decay of waste, while the temperature of the host rock is generally in the regime of 50°C. Indeed, tests (Jansen *et al.* 1983) on commercial waste forms in the Palo Duro Basin of the United States show a temperature peak of 220°C after 1 year. After the initial period of 1–10 years, the temperature is expected to decrease exponentially with time as radioactivity wanes. The primary effect of the initial heating of the barrier is the loss of moisture from the annular region that is proximal to the waste canister. Barrier mix proportioning tests for the Stripa Mine repository (Pusch 1994) indicate that a typical barrier backfill would be compacted to a dry density of 1.8 g/cm^3 and moisture content of about 9%. Under such high densities, more than 95% of the moisture exists in interlamellar spaces in clay.

While heat generation tends to extend the zone of desiccation radially outwards, the intense hydrostatic pressure in the host rock, which for a repository in the depth range of 600–1128 m may reach 3.0 MPa for tuff and 11.34 MPa for basalt (Kehrman *et al.* 1987), tends to push moisture into the barrier. Thus, following waste embedment, ion-laden water is likely to infiltrate portions of the barrier and may eventually gain access to the surface of the waste canister. However, access of moisture to the waste canister may not necessarily produce significant transport of radionuclides from the canister if the barrier maintains its integrity and the radioactive decay functions of the radionuclides are such that insignificant quantities of the substances are available for transport. Essentially, during the barrier system infiltration phase, the location of the infiltration front within the barrier will depend on the physical equilibrium with the outward moving desiccation front induced by temperature increase. The decay of temperature with time implies that the infiltration will reach the canister surface (full barrier saturation) eventually. Figure 1 is an illustration of the conceptual relationship between canister surface temperature and barrier moisture content at two different stages of barrier system service. The sequence of infiltration-related phenomena is important because it establishes the relative time for the access of potentially corrosive mineralized water to the canister, a requirement for the onset of canister attack phenomena, which may eventually lead to the release of radionuclides from the disposal system into the host rock.

The heat generation phase precedes the heat decay phase. The former may last for about 10 years based on the radioactive decay pattern of wastes that are currently in temporary storage in the United States. The heat decay phase may last for hundreds of years thereafter. A transition zone, which is prominent during the heat generation phase, separates the outer saturated zone from the inner dehydrated zone. In the dehydrated zone, the moisture content is much below the initial moisture content of the barrier. Relative to the other two zones, there is a high moisture content gradient across the transition zone.

A temperature gradient exists across the barrier at any given time during the heat generation and decay phases. The highest temperature is at the canister surface and the lowest temperature is at the rock/barrier interface. As time progresses, the barrier temperature is expected to fall. Consequently, the combined effect of hydrostatic and thermal transport displaces the saturation front inward. In this physical model, the challenge is quantitatively to demarcate the saturated and unsaturated zones in time and space within the barrier. A transition zone characterized by dissolution, leaching and precipitation of minerals would separate the saturated and unsaturated zones. Under the expected high temperature conditions around the canister, at least initially, smectite crystals may dissolve in the inner portion of the transition zone of the barrier. In this zone, high temperature that is coupled with adequately high moisture content (as vapour, and below saturation), promotes the release of silicon (Si^{4+}) and aluminium (Al^{3+}) from smectites. These cations may diffuse outward. Silicon (Si^{4+}) is expected to precipitate in the colder portion of the barrier as amorphous silica or quartz within the clayey barrier matrix. Salts from the host rock may also precipitate in the transition zone.

Comprehensive modelling of the long-term performance of repositories in terms of the quantity and rate of release of radionuclides requires the estimation of concentration gradient of radionuclide and moisture, temperature, flow velocity, electric potential and pressure; and the coupled processes that complicate relevant analyses. Precisely, barrier

alteration influences the accuracy of flux estimates in two regards: zones with different textural (and hence flow) characteristics develop in the barrier such that the assumption of a single-layered homogeneous barrier in models becomes invalid; and for a given barrier thickness, the relative thickness of the zones that develop change with time such that the accuracy of single spatial compartment models is compromised. Conductive heat flow, liquid infiltration and radionuclide diffusion are the principal flow processes that are likely to be influenced by any significant change in relevant barrier material properties.

4. Modelling of illitization

Rossi *et al.* (2003) describe a simple diffusion model for the evolution of moisture and temperature in the barrier material

$$(\rho c + w\rho_w c_w)\frac{\partial T}{\partial t} - \nabla[K(w)\nabla T] = -\rho_w L_v\left[Z_e w e^{-L_v/RT} - Z_c V_{eq}\right] \quad (3)$$

$$\frac{\partial w}{\partial t} - \nabla[D(w)\nabla w] = -Z_e w e^{-L_v/RT} + Z_c V_{eq} \quad (4)$$

The essentials of saturation-dependent moisture transport can be found in Philip (1996). Appropriate units for all parameters used in these simulations are presented in the Appendix. The dependent variables T and w represent the temperature and volumetric moisture content, respectively. Equation (3) represents conservation of thermal energy; equation (4) is the Richards' equation for unsaturated flow, modified to account for evaporation and condensation. The parameter ρ is the density of clay, ρ_w is the moisture density, c is the specific heat of the barrier, c_w is the specific heat of water, L_v is the latent heat of vaporization of the water, Z_e is the rate constant associated with the Arrhenius model for evaporation, R is the universal gas constant, Z_c is the condensation rate and V_{eq} the equilibrium vapour content in the porous matrix, assumed constant. The thermal diffusivity $K(w)$ is given by

$$K(w) = k_b + wk_w \quad (5)$$

where k_b is the thermal conductivity of the barrier material and k_w is the thermal conductivity of water. $K(w)$ measures heat flux per unit area per unit thermal gradient. In equation (5), the thermal conductivity of water k_w must be multiplied by the volumetric water content w to account for the fact that water does not entirely fill the region under consideration. No such correction is needed for the thermal conductivity of the barrier material because porosity is already accounted for in the measured value of k_b. The equilibrium vapour content V_{eq} is determined from the far-field temperature (T_0) and moisture content (w_0) so that condensation rate is equal to the evaporation rate:

$$V_{eq} = \frac{Z_e w_0 e^{-L_v/RT_0}}{Z_c} \quad (6)$$

Thermally driven moisture transport is not include, as discussed in Philip and DeVries (1957) because the diffusivity coefficient is five orders of magnitude less than $D(w)$. The hydraulic diffusivity $D(w)$ is fitted to measured data on Yolo light clay provided by Philip (1996):

$$D(w) = 10^{-d(w)}\ \mathrm{m^2/s} \quad (7)$$

where

$$d(w) = \begin{cases} 8.3 + 15000(w-0.03)^2, & w < 0.03 \\ 8.3 - 25171.7(w-0.1)(w-0.03)^2, & w \in [0.03, 0.080825) \\ 7.0 + 25(\mathrm{w}-0.4)^2, & w \in [0.080825, 0.4) \\ 7.0 + \dfrac{(w-0.4)^2}{4(w-0.3)(w-0.5)}, & w \in [0.4, 0.5) \end{cases} \quad (8)$$

Using estimates based on information extracted from figure 2, the hydraulic diffusivity for illite is taken to be $1.5D(w)$.

In the present paper, the novelty is the consideration of the effects of illitization on the barrier system. Assuming that the illitization process follows a kinetic pathway of order α as suggested by Pytte (1982) and Huang *et al.* (1993), then

$$\frac{\mathrm{d}S}{\mathrm{d}t} = -\kappa S^{\alpha}[K^+]^{\beta} \quad (9)$$

where S denotes the mole fraction of smectite in the barrier material, $[K^+]$ is the potassium ion concentration and β is the

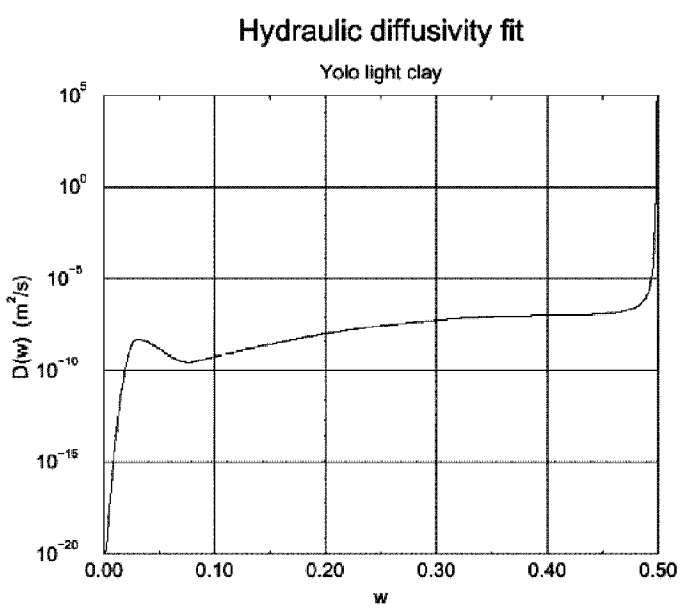

Figure 2. Hydraulic diffusivity for Yolo light clay.

potassium order parameter. κ is the Arrhenius-type illitization reaction rate:

$$\kappa = Ae^{-E_a/RT} \tag{10}$$

In equation (10), A is the pre-exponential factor and E_a is the activation energy for the reaction. Since the flow is unsaturated and the conversion reaction takes place only where water is present, the term on the right side of equation (9) is multiplied by w in this paper. If b denotes the potassium ion concentration per unit volume of space, then $[K^+] = b/w$ and the smectite conversion equation can be expressed as

$$\frac{\partial S}{\partial t} = -Ae^{-E_a/RT} S^{\alpha} \left(\frac{b}{w}\right)^{\beta} w \tag{11}$$

As in Huang *et al.* (1993), the values $\alpha = 2$ and $\beta = 1$ are used in the current paper. The conservation of potassium ions is modelled by the equation

$$\frac{\partial b}{\partial t} + \nabla[-bD(w)\nabla w] = \sigma\nabla^2 b - 11.6Ae^{-E_a/RT} S^{\alpha} \left(\frac{b}{w}\right)^{\beta} w \tag{12}$$

Here $\nabla[-bD(w)\nabla w]$ gives the convection of potassium ions by water, $\sigma\nabla^2 b$ gives the potassium ion diffusion, and the final term gives the depletion of potassium ions in the conversion reaction as determined by the stoichiometric equation (1). Values of all parameters are listed in the Appendix.

The effects of illitization on the diffusion coefficients D, K and σ and the specific heat c are modelled by assuming that each of these quantities is a weighted average of its value in smectite and its value in illite

$$D(w) = SD_S(w) + (1 - S)D_I(w) \tag{13}$$

$$K(w) = SK_S(w) + (1 - S)KE(w) \tag{14}$$

$$c = Sc_S + (1 - S)cE \tag{15}$$

$$\sigma = S\sigma_S + (1 - S)\sigma_I \tag{16}$$

(D_S is given by equation (7); $D_I = 1.5D_S$.) As in Rossi *et al.* (2003), it is assumed that all dependent variables depend only on radial distance from the central axis of the waste canister (r) and time (t). The equations for temperature, moisture content, smectite mole fraction and potassium ion concentration then reduce to the following four equations:

$$(\rho c + w\rho_w c_w)\frac{\partial T}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left[rK(w)\frac{\partial T}{\partial r}\right] - H_N(T - T_0) - \rho_w L_v(Z_e w e^{-L_v/RT} - Z_c V_{eq}) \tag{17}$$

$$\frac{\partial w}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left[rD(w)\frac{\partial w}{\partial r}\right] - Z_e w e^{-L_v/RT} - Z_c V_{eq} \tag{18}$$

$$\frac{\partial S}{\partial t} = -Ae^{-E_a/RT} S^2 b \tag{19}$$

$$\frac{\partial b}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left[rbD(w)\frac{\partial w}{\partial r}\right] + \sigma\left[\frac{\partial^2 b}{\partial r^2} + \frac{1}{r}\frac{\partial b}{\partial r}\right] - 11.6Ae^{-E_a/RT} S^2 b \tag{20}$$

To supplement equations 13 to 20, it is necessary to specify appropriate boundary and initial conditions. As discussed in Rossi *et al.* (2003), the Newton cooling term $H_N(T\text{-}T_0$ in the temperature equation is a modification specific to the axisymmetric problem. In effect, it replaces a boundary condition at the top and bottom of the repository. The radius of the waste canister is R_0, and the engineered clay barrier occupies the space $R_0 < r < R_1$. Thus, the spatial domain for the solution of the model equations is $R_0 < r < R_1$. There is a specified thermal flux H_c from the waste canister at R_0, and it is assumed that the moisture and potassium ions in the clay barrier cannot penetrate the canister. Furthermore, it is assumed that the temperature, moisture fraction, and potassium ion concentration at the outer edge of the clay barrier have the constant 'far field' values T_0, w_0, and b_0 respectively. Thus, the boundary conditions are

$$\frac{\partial T}{\partial r}(R_0, t) = -\frac{H_c e^{-\lambda t}}{2\pi R_0(k_b + wk_w)} \tag{21}$$

$$\frac{\partial w}{\partial r}(R_0, t) = 0 \tag{22}$$

$$\frac{\partial b}{\partial r}(R_0, t) = 0 \tag{23}$$

$$T(R_1, t) = T_0 \tag{24}$$

$$w(R_1, t) = w_0 \tag{25}$$

$$b(R_1, t) = b_0 \tag{26}$$

For initial conditions, it can be assumed that the temperature, moisture, and potassium ion concentration are everywhere equal to the ambient values and that the clay is 100% smectite:

$$T(r, 0) = T_0 \tag{27}$$

$$w(r, 0) = w_0 \tag{28}$$

$$S(r,0) = 1 \tag{29}$$

$$b(r,0) = b_0 \tag{30}$$

Equations (17) to (20) are solved numerically using the MATLAB routine *pdepe*. This routine implements the method of lines, in which the spatial domain is discretized and the resulting coupled system of ordinary differential equations is solved using standard numerical techniques. Computational experiments were performed with an outer ring of host rock, but these exhibited no significant differences from the experiments presented in the present paper.

5. Analysis of results

To gain some insight into the general behaviour of the simplified model presented here, three simulations of fluid flow with K^+ ions into an earthen barrier of a nuclear waste repository were performed. The first simulation is a reference solution where the illitization reaction rate is zero. The second simulation includes illitization, but the properties of the illite are considered identical with those of smectite. Essentially, the smectite acts as a potassium sink. Finally, the third simulation includes illitization where the illite parameter values (specific heat, thermal diffusivity, hydraulic diffusivity and potassium ion diffusivity) differ from the smectite parameter values by 50%. (See the Appendix for the parameter values used.) An additional calculation was performed in which the illite parameter values differed from smectite parameter values by 75% and 100%, but found no observable differences from the 50% calculation. From these calculations, the following observations are made.

1. The vaporization front acts as a transport mechanism for potassium as well as fluid that moves from the far field through the barrier to the canister. In figure 3, the potassium concentrations at the container are estimated as a function of time when there is no mineralogical conversion of smectite to illite (illitization). Diffusion suppresses the concentration gradient created by the vaporization and transport of fluid at the front. The dramatic difference is shown in figure 4 which shows the spatial variation in potassium concentration after 1000 years. However, it should be noted that convective transport is effective over great distances over long periods of time while diffusive transport is effective over short distances in short periods of time. In figure 4, the temporal evolution of potassium near the canister is observed.
2. Figures 5, 6 and 7 show the spatial variation of the other dependent variables after 1000 years. The effect of illitization on the temperature and moisture content is seen to be very small.

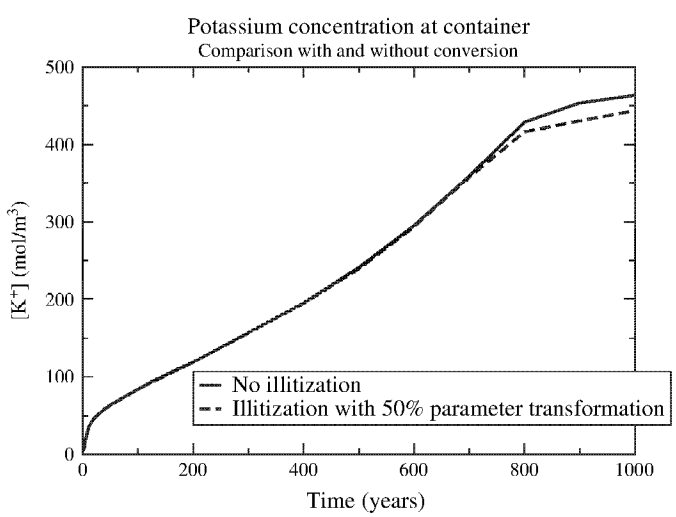

Figure 3. $[K^+]$ concentration at the container wall versus time.

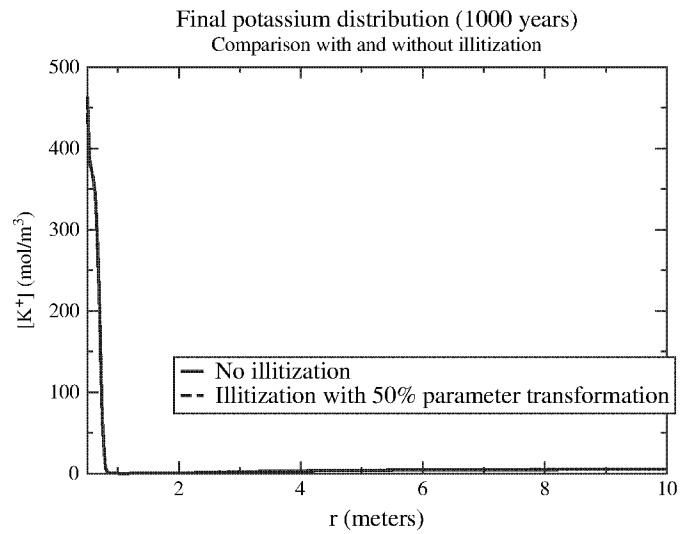

Figure 4. $[K^+]$ concentration after 1000 years (solid curve overlaps dotted curve).

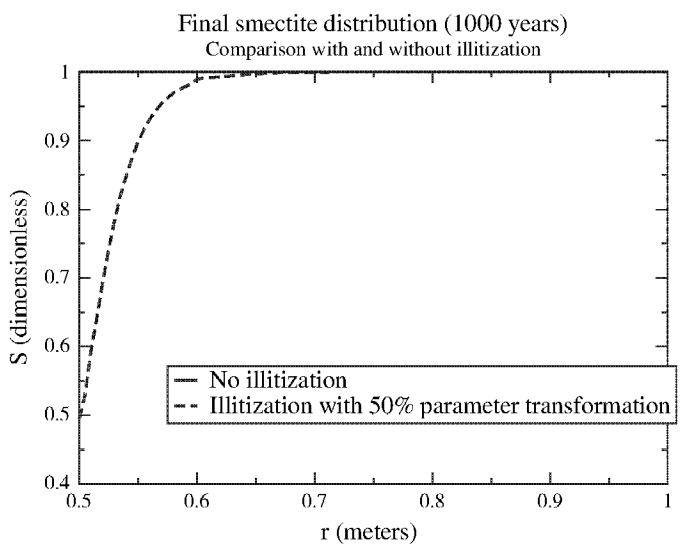

Figure 5. Smectite distribution after 1000 years (the solid curve is the horizontal line $S = 1$).

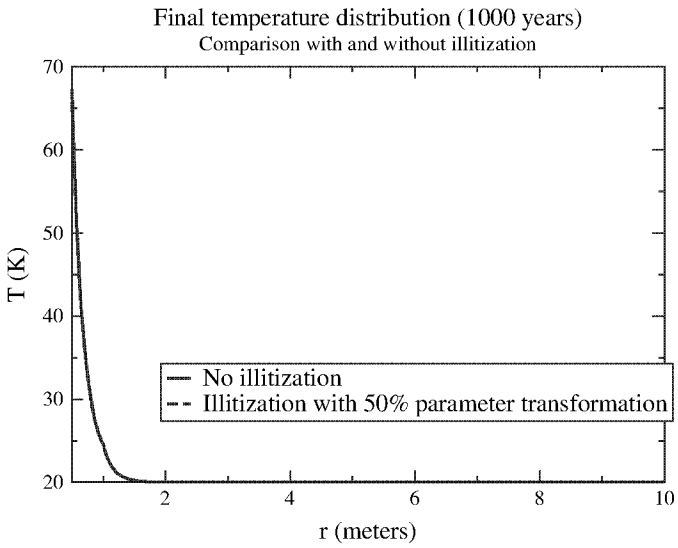

Figure 6. Spatial variation of temperature after 1000 years (solid curve overlaps dotted curve).

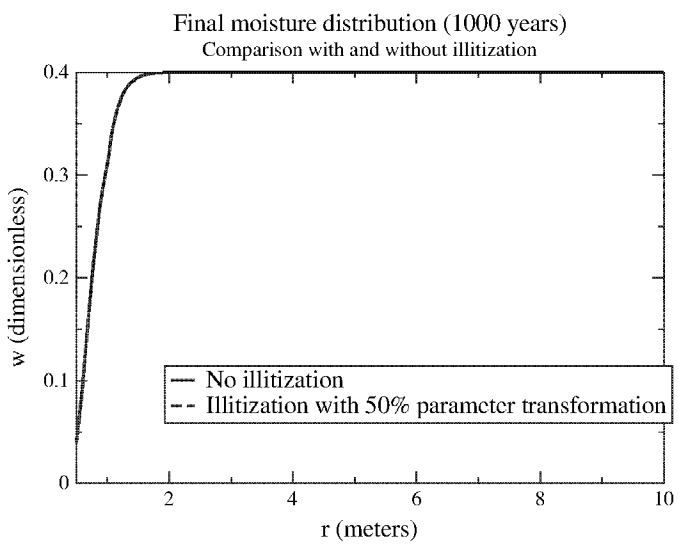

Figure 7. Spatial variation of moisture content after 1000 years (solid curve overlaps dotted curve).

6. Conclusions

The numerical simulations described in the present paper indicate that coupling of fluid transport with reaction front processes can provide some insight into the rate of accumulation of excess ions on the surfaces of deeply buried radioactive waste canisters. This is necessary to estimate the potential for significant corrosion of the waste canister, a development that could lead to release of available radionuclides into a clay barrier system.

Based on the simulations that are performed using assumed but realistic and typical magnitudes of parameters that determine fluid and heat transport, reaction rates, barrier texture, dimensions and configuration, it is observed that illitization, a key potential barrier transformation mechanism is unlikely to lead to significant accumulation of excess potassium (K^+) ions at the surface of the waste-containing canister.

Within an analytical time-frame of up to 1000 years, the difference between the quantities of K^+ at the canister for an illitized barrier and non-illitized barrier does not exceed 25 mol/m^3. For the parameter magnitudes used in the simulation, the distribution of K^+ is such that a narrow ($\approx$0.4 m) rim that has a maximum K^+ concentration in pore solution of about 450 mol/m^3 develops.

These results reflect the fact that temperature across the barrier is only high enough within a narrow zone of less than 1 m around the heat-generating canister to induce significant illitization of the barrier.

References

Bethke, C. M., Vergo, N. and Altaner, S. P., Pathways to smectite illitization. *Clay Clay Miner.*, 1986, **34**, 125–135.

Boles, J. R. and Franks, S. G., Clay diagenesis in Wilcox sandstones of southwest Texas: Implications of smectite diagenesis on sandstone cementation. *J. Sed. Petrol.*, 1979, **49**, 55–70.

Carnahan, C. L., Inertial effects in transport of radioactive waste through engineered barriers. *Proceedings of Symposium of Scientific Basis for Nuclear Waste Management*, VI, Boston, Massachusetts, USA, 1982, pp. 745–752.

Carr, J. A. and Basham, S. J., Waste package conceptual designs for salt. In *Waste Management '83: Proceedings of the Symposium on Waste Management*, edited by R. G. Post, Vol. II, 1983, p. 17–20.

Chapman, N. A., McKinley, I. G. and Smellie, J. A. T., The potential of natural analogues in assessing systems for deep disposal of high-level radioactive waste. TR 84-16, SKB, Stockholm, Sweden, 1984.

Chen, J., Anandarajah, A. and Inyang, H. I. Pore fluid properties and compressibility of kaolinite. *ASCE J. Geotech. Geoenviron.l Engng*, 2000, **16**(9), 768–807.

Daniels, J. L., Inyang, H. I. and Iskandar, I., Durability of Boston Blue Clay in waste containment applications. *ASCE J. Mater. Civil Engng*, 2003, **15**, 144–152.

Eberl, D. D. and Hower, J., Kinetics ofillite formation. *Geol. Soc. Amer. Bull.*, 1976, **87**, 1326–1330

Fang, H. Y., Daniels, J. L. and Inyang, H. I., Enviro-geotechnical considerations in waste containment system design and analysis. *Proceedings of International Containment Technical Conference and Exhibition*, St. Petersburg, Florida, 1997, pp. 414–420.

Fleming, L. N. and Inyang, H. I., The permeability of clay-modified flyash under thermal gradient. *J. Mater. Engng*, 1995, **7**, 178–182.

Huang, W-L., Longo, J. M. and Prevear, D. R., An experimentally derived kinetic model for smectite-to-illite conversion and its use as a geothermometer. *Clay Clay Miner.*, 1993, **41**, 162–177.

Inyang, H. I. and Tumay, M. T., Containment systems for contaminants in the subsurface. In *Encyclopedia of Environmental Control Technology*, pp. 175–215, 1995 (Gulf Publishing Company: New York).

Inyang, H. I., Iskandar, A. and Parikh, J. M., Physico-chemical interactions in waste containment barriers. *Encyclopedia of Environmental Analysis and Remediation*, 1998a, Vol. 2. pp. 1158–1165 (John Wiley: New York).

Inyang, H. I., Fang, H. Y., Iskandar, A. and Choquette, M. R., Chemical and mineralogical analyses of clayey barrier materials. *Encyclopedia of Environmental Analysis and Remediation*, 1998b, Vol. 8, pp. 5131–5141 (John Wiley: New York).

Inyang, H. I., A Weibull-based reliability analysis of waste containment system. *Proceedings of First International Congress on Environmental Geotechology*, Alberta, Canada, 1994, pp. 273–278.

Inyang, H. I., Balancing prescriptive and performance-based design specifications for waste containment systems. *ASCE J. Environ. Engng*, 1999, **125**, 491.

Jansen, G., Raines, G. E. and Kircher, J. F., Performance analysis of conceptual waste package designs in salt repositories. *Proceedings of Conference on Scientific Basis for Nuclear Waste Manage. VII*, Boston, Massachusetts, USA, 1983. pp. 445–454.

Kanno, T., Fujita, T., Takeuchi, S., Ishikawa, H., Hara, K. and Nakano, M., Coupled thermo-hydro-mechanical modeling of bentonite buffer material. *Proceedings of the International Workshop on Hydro-thermo-Mechanics of Engineered Clay barriers and Geological Barriers*, Montreal, Canada, 5–7 July 1995, McGill University, pp. 1–50.

Kargbo, D. M., Fanning, D. S., Inyang, H. I. and Duell, R. W., The environmental significance of acid sulphate clays as waste covers. *Environ. Geol.*, 1993, **22**(3), 218–226.

Kehrman, R. F., Schornhorst, J. R., Kelley, J. L. and Bolmgren, C. R., Conceptual waste package designs for disposal of high-level nuclear wastes in geologic repositories. In *The Technology of High-Level Nuclear Waste Disposal*, edited by P. L. Hofmann, 1987, pp. 22–45 (US Department of Energy: Washington, D.C).

Philip, J. R. and de Vries, D. A., Moisture movement in porous materials under temperature gradients. *Trans. Amer. Geophys. Union*, 1957, **38**, 222–232.

Philip, J. R., Mathematical physics of infiltration on at and sloping topography. In *Environmental studies: Mathematical, Computational, and Statistical Analysis*, Vol. 79, pp. 327–349, 1996 (Springer-Verlag: New York) pages 327–349.

Pusch, R., Waste disposal in rock. *Developments in Geotechnical Engineering*, Vol. 76, 1994 (Elsevier Publishing: New York).

Pytte, A., The kinetics of the smectite to illite reaction in contact metamorphic shales. M. A. Thesis, Dartmouth College, Hanover NH, 1982.

Reddi, L. N. and Inyang, H. I., *Geoenvironmental Engineering: Principles and Applications*, 2000, pp. 494 (Marcel Dekker: New York).

Rossi, L., Inyang, H. I., Graham-Eagle, J. and Pennell, S., A model of coupled heat and moisture transport in an annular clay barrier. *ASCE J. Environ. Engng*, 2003, **130**, 855–862.

Seitz, R. R., Sagar, B. and Davis, J. D., Numerical modeling of radionuclide transport through radionuclide seals. *Proceedings of Symposium on Waste Management*, Tucson, Arizona, USA, 1987, pp. 637–644.

Soudek, A., Jahnke, P. M. and Radke, C. J., Ion exchange equilibria and diffusion in engineered backfill. *Proceedings of NRC Conference on Nuclear Waste Geochem*, NUREG/CP-0052, Reston, Virginia, USA, 1983, pp. 172–203.

Whitney, G., Role of water in the smectite-to-illite reaction. *Clays Clay Miner.*, 1990, **38**, 343–350.

APPENDIX

Notation

Independent variables

r	radial distance from centre of waste canister (m)
t	time (s)

Dependent variables

b	number of moles of potassium ions per unit volume of space (moles/m^3)
S	mole fraction of smectite in barrier (dimensionless)
T	temperature of water and matrix (K)
w	volumetric moisture content (dimensionless)

Physical parameters

The following physical constants are used in the computations presented in the present paper. Some have been obtained from standard tables. Others have been determined from studies such as those of Carr and Basham (1983), Huang *et al.* (1993), and Kanno *et al.* (1995). Parameter values for the host rock are assumed to be the same as the values for smectite unless otherwise noted.

A	pre-exponential factor in smectite conversion equation (1000 m^3/mole s)
c_I	specific heat of illite (10^3 J/kg K)
c_S	specific heat of smectite (10^3 J/kg K)
c_w	specific heat of water (4.19×10^3 J/kg K)
E_a	activation energy of illitization reaction (1.17×10^5 J/mole)
H_c	thermal output of nuclear waste (2.2×10^3 W)
H_N	thermal transfer coefficient (20 W/m^3 K)
k_b	thermal conductivity of barrier material (0.33 W/m K)
k_I	thermal conductivity of illite (1.44 W/m K)
k_S	thermal conductivity of smectite (0.33 W/m K)
k_w	thermal conductivity of water (0.67 W/m K)
L_v	heat of vaporization of water (4.07×10^4 J/kg)
R	universal gas constant (8.31 J/mole K)
R_0	radius of storage vessel (0.5 m)
T_0	ambient temperature underground (20 K)
w_0	ambient moisture fraction (0.4)
Z_c	condensation rate (4.7 s^{-1})
Z_e	Arrhenius evaporation rate (3.41×10^4s^{-1})
ρ	density of porous matrix (1.6×10^3 kg/m^3)
ρ_w	density of water (10^3 kg/m^3)
σ	diffusivity of K^+ ions in water (2×10^{-10} m^2/s)

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